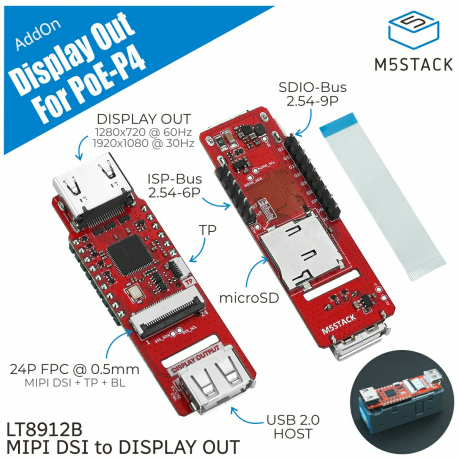
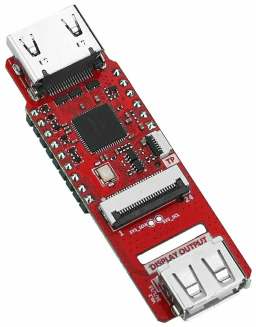
AddOn Display Out For PoE-P4

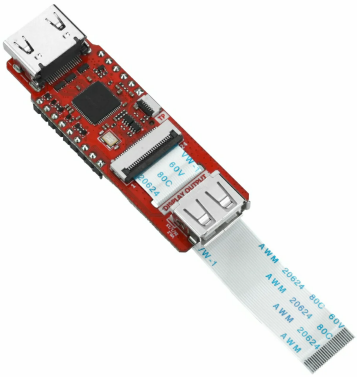
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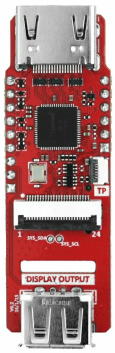


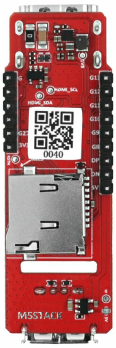










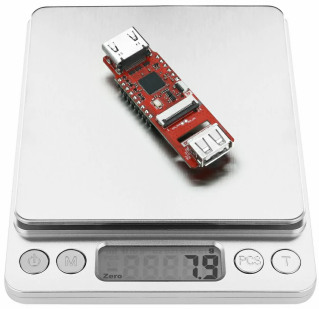












Description

AddOn Display Out For PoE-P4 is a display output expansion module designed for Unit PoE-P4. Based on the `LT8912B` chip, it converts the `MIPI DSI` display signal from Unit PoE-P4 into `HDMI` output, making it easy to connect an external monitor for visualizing local graphical interfaces, status information, and control screens.

In addition to display output, the module integrates expansion capabilities for touch and local storage. The onboard `TP` interface connects to a touch controller and provides the required `I2C` communication, reset, and interrupt signals for touch input and UI interaction. The onboard microSD slot connects to the main controller's `SDIO` bus and can store images, UI resources, configuration files, or other display-related data, making it easier to build a more complete local display solution. The module also exposes a `USB Type-A` interface for connecting peripherals such as a mouse or keyboard. It is suitable for PoE display terminals, industrial HMI, information dashboards, embedded control panels, and similar applications.

Note

Display Compatibility

When using AddOn Display Out For PoE-P4 with an external display, ensure that the display supports the output resolution and refresh rate currently configured on the device. Otherwise, the display may not function properly.

Features

- Display output expansion module for Unit PoE-P4
- Converts `MIPI DSI` to `HDMI` display output based on `LT8912B`
- Standard `HDMI Type-A` interface for quick connection to external displays
- Onboard `TP` touch interface, supporting touch input and UI interaction expansion
- Onboard microSD slot connected to the main controller's `SDIO` bus, supporting local storage of images, UI resources, and configuration files
- Onboard `USB Type-A` interface, supporting peripherals such as mouse and keyboard
- Supports building an integrated display solution combining display, touch, and storage
- Supports two output resolutions: `1280 x 720 @ 60Hz` and `1920 x 1080 @ 30Hz`

Includes

- 1 x AddOn Display Out For PoE-P4
- 1 x 24P FPC Cable (0.5mm pitch)

Applications

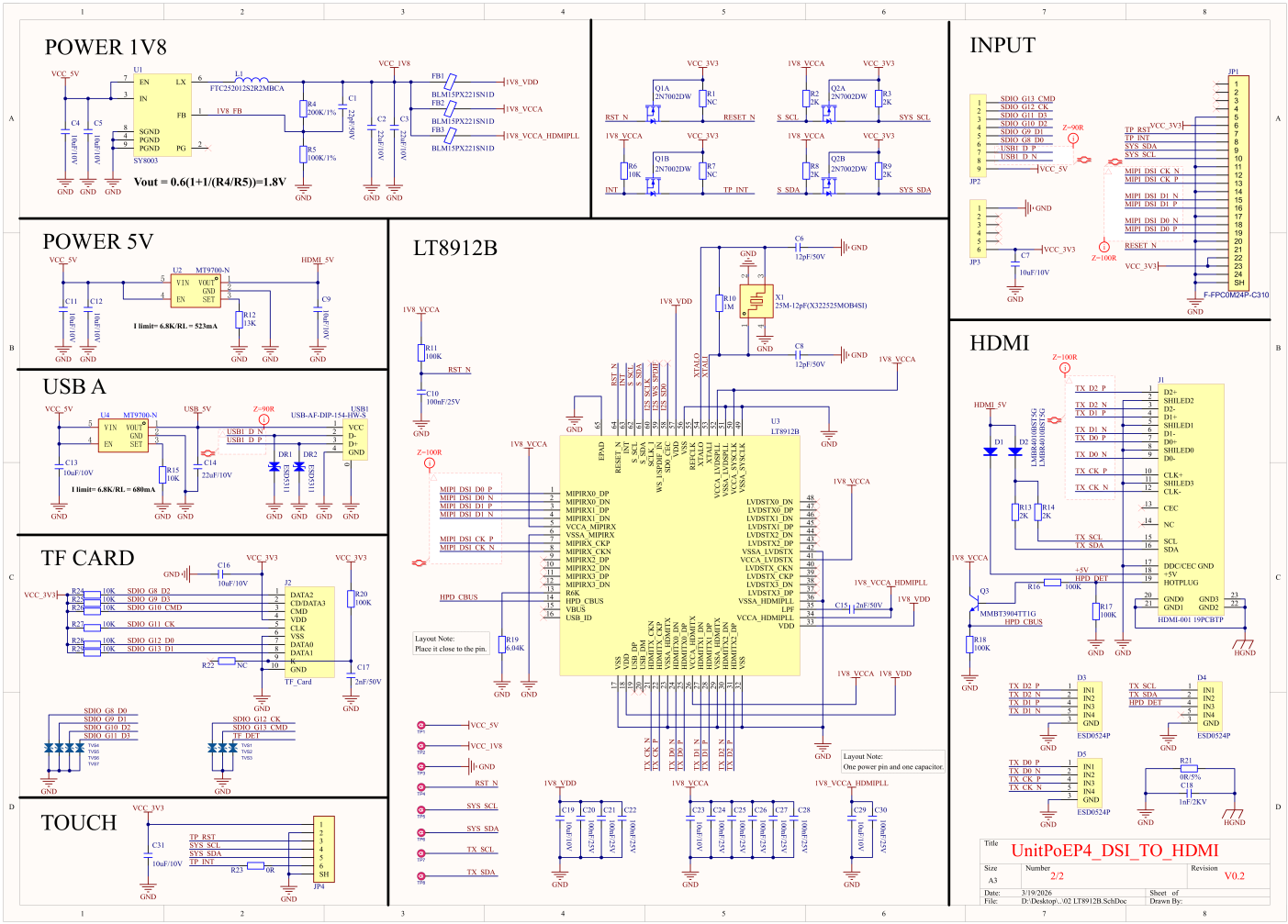
- PoE display terminal
- Industrial HMI
- Device status visualization dashboard
- Human-machine interaction control panel with touch display
- Local display and debugging for embedded devices

Specifications

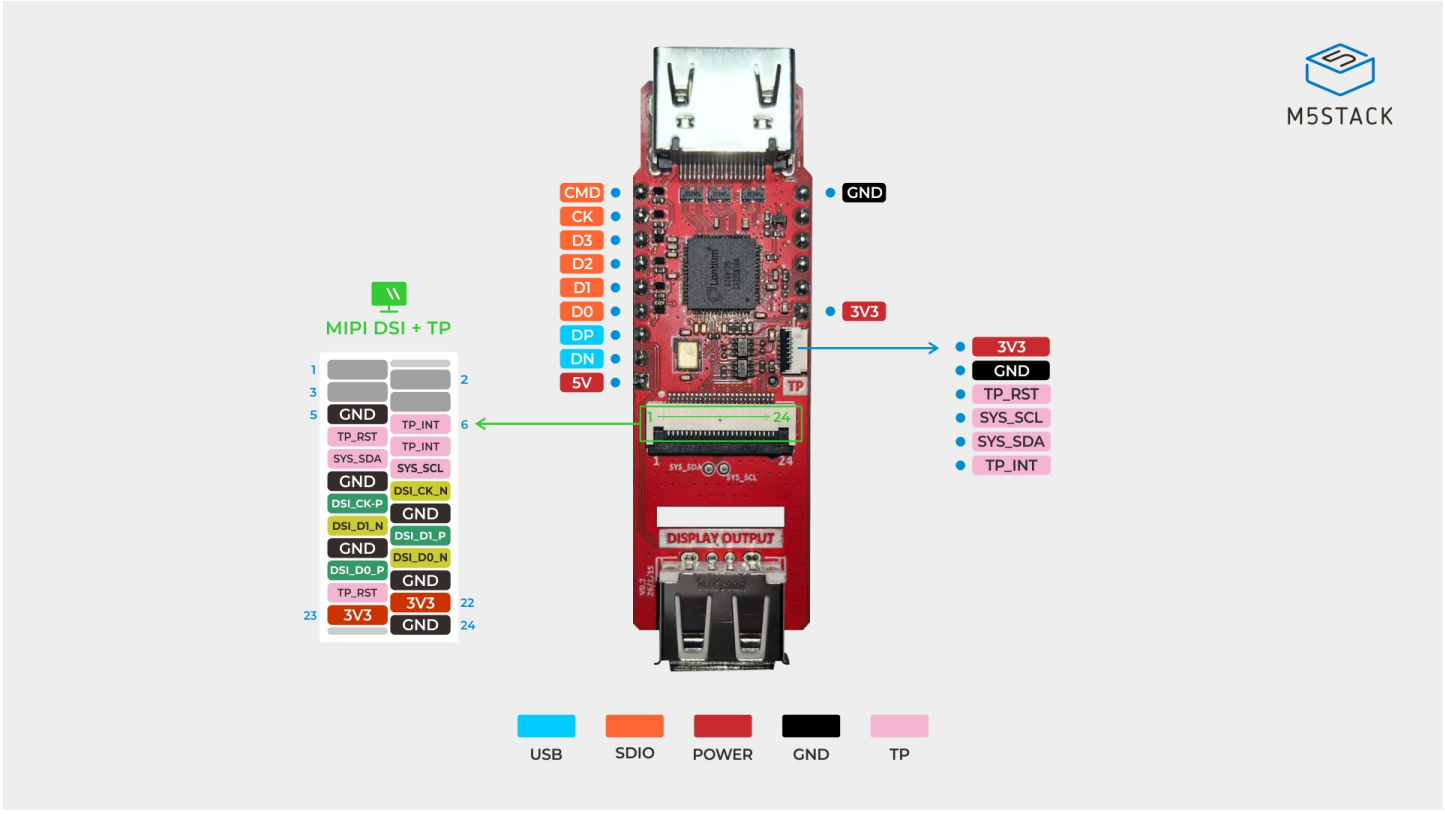
Specification	Parameter
Conversion Chip	LT8912B
Main Input Interface	24P FPC (0.5mm pitch)
Input Signal	MIPI DSI 2-Lane / SYS_SDA / SYS_SCL / TP_INT / TP_RST / USB1_DP / USB1_DN / SDIO / RESET_N
Video Output Interface	1 x HDMI Type-A
Touch Expansion Interface	1 x 0.55-6P Touch interface
USB Expansion Interface	1 x USB Type-A
USB Load Capacity	5V @ 0.5A
Storage Expansion	1 x microSD slot
Power Input	DC 5V (powered by the Unit PoE-P4 expansion interface)
Display Output Timing	1280 x 720 @ 60Hz / 1920 x 1080 @ 30Hz
Standby Power Consumption	DC 5V@74.55mA (whole unit)
Operating Power	DC 5V@186.83mA (whole unit)
Product Size	20.3 x 62.9 x 15.5mm
Product Weight	7.9g
Package Size	138.0 x 93.0 x 14.0mm
Gross Weight	11.2g

Schematics

- AddOn Display Out For PoE-P4 Schematics PDF



PinMap



24P FPC Input Interface

Signal Group	Corresponding Signal
TP (I2C)	TP_RST, TP_INT, SYS_SDA, SYS_SCL
MIPI DSI	MIPI_DSI_CK_P/N, MIPI_DSI_D0_P/N, MIPI_DSI_D1_P/N
Control / Power	RESET_N, 3V3, GND

6P Touch Interface

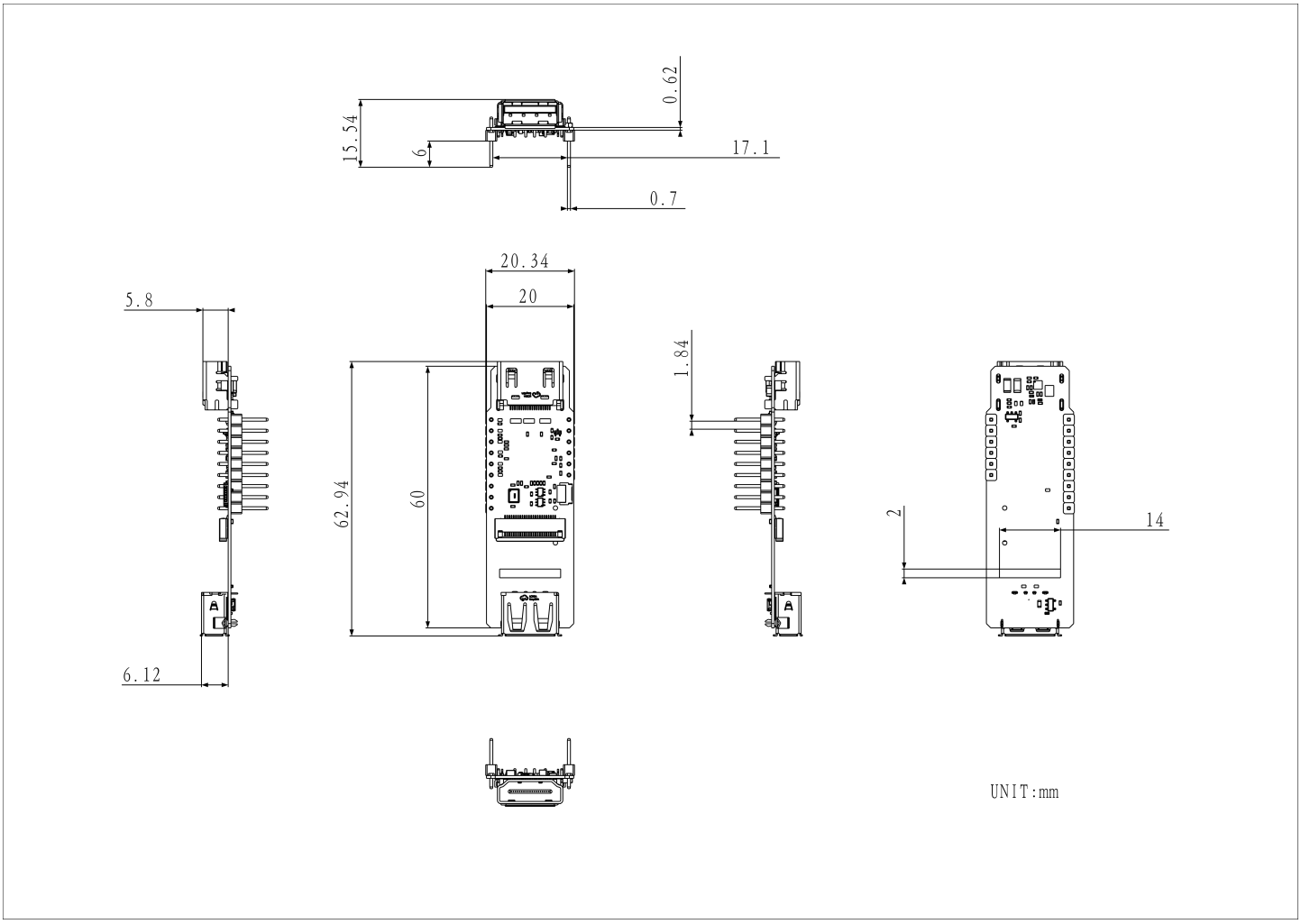
Interface	Corresponding Signal
6P Touch	3V3, TP_RST, SYS_SCL, SYS_SDA, TP_INT, GND

SDIO & USB & POWER

PIN	LEFT	RIGHT	PIN
SDIO_CMD	1	1	GND
SDIO_CK	2	2	
SDIO_D3	3	3	
SDIO_D2	4	4	
SDIO_D1	5	5	
SDIO_D0	6	6	3V3
USB1_DP	7		
USB1_DN	8		
5V	9		

Model Size

- AddOn Display Out For PoE-P4 Model Size PDF



Softwares

Arduino

- AddOn Display Out For PoE-P4 Arduino Example

UiFlow2

- AddOn Display Out For PoE-P4 UiFlow2 Docs

Video

- AddOn Display Out For PoE-P4 product introduction and feature demonstration

U221_AddOn_Display_Out_For_PoE-P4_video_EN.mp4